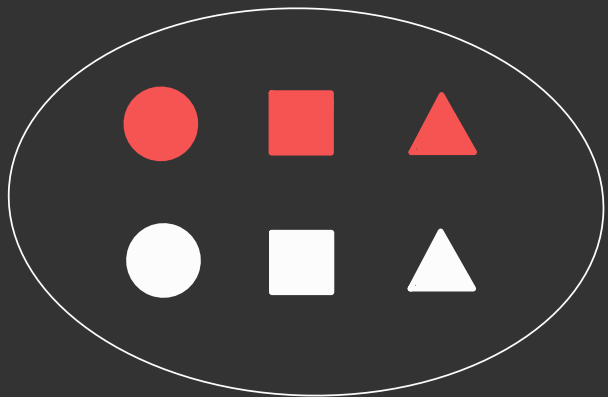


conditional probability

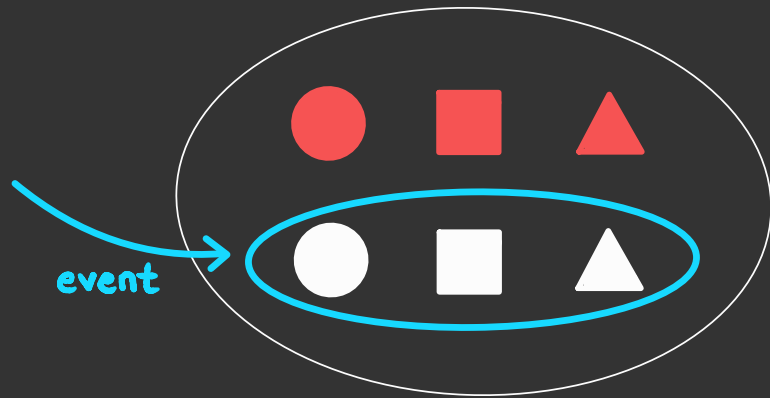
CSCI
270

if we draw some
kind of misfortune,
what is the probability
it isn't "great"?



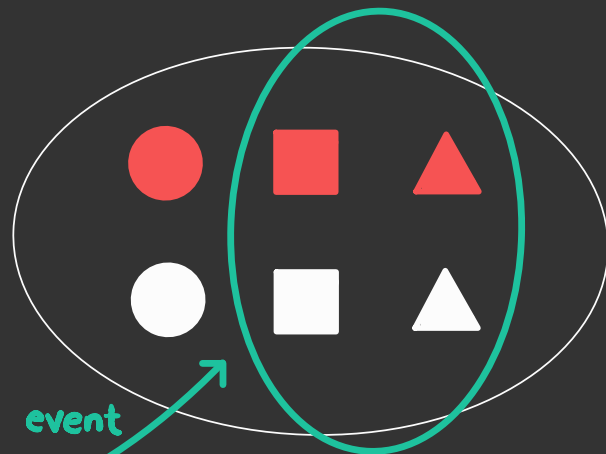
sample space Ω
(set of possible outcomes)

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sample space Ω
(set of possible outcomes)

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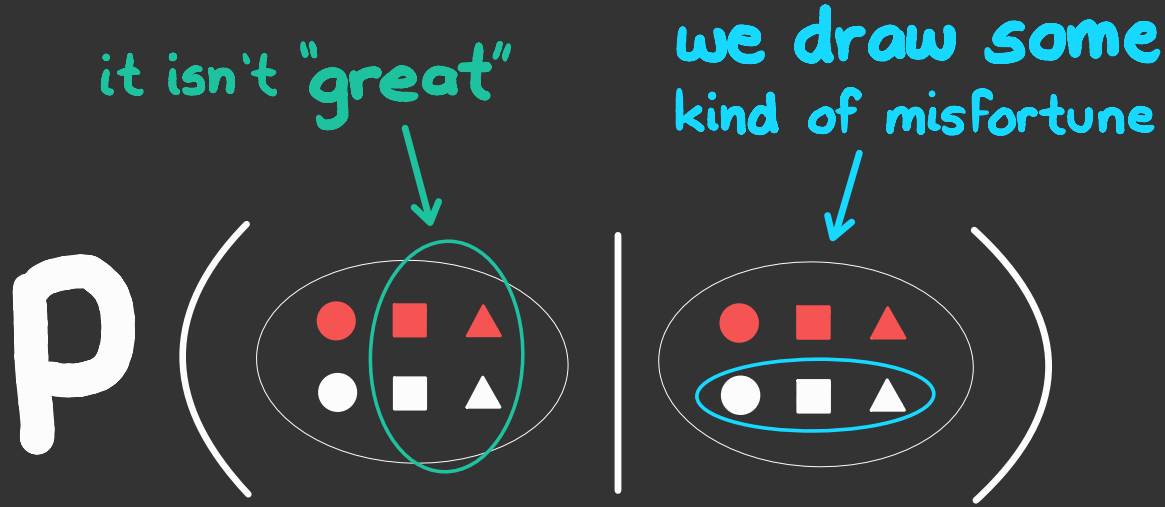


event

sample space

(set of possible outcomes)

Ω



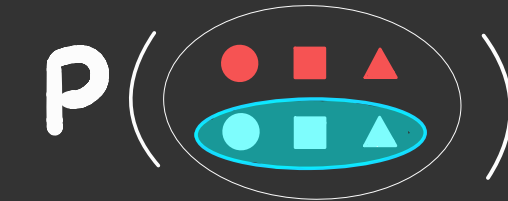
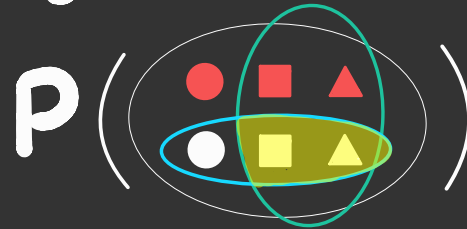
"the probability it isn't great, given we draw some kind of misfortune"

conditional probability

let A, B be events

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

e.g.



exercise



given the definition of conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

show that:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

exercise

given the definition of conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

show that:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

from the definition of conditional probability :

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{and} \quad P(B|A) = \frac{P(B \cap A)}{P(A)}$$

rearranging, we get:

$$P(A|B)P(B) = P(A \cap B) \quad \text{and} \quad P(B|A)P(A) = P(B \cap A)$$

the right-sides are the same, so:

$$P(A|B)P(B) = P(B|A)P(A)$$

rearranging, we get:

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rearranging, we get:

bayes
rule



$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

let A, B be events

conditional probability

$y|x$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

bayes rule



$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$